LAB 6

CODE AND BRANCH COVERAGE ANALYSIS

Marathon Registration Application

Student Name	Aditya R Badde
SRN	PES2UG25CS802
Section	A
Course / Institution	PES University

GitHub Repository: https://github.com/adityabadde/L6_L7_60_marathon_registration

1. Objective

The objective of this lab is to understand the difference between code coverage and branch coverage, use a coverage tool to measure test completeness, identify untested execution paths, and write additional test cases for edge cases and decision branches.

2. Application Overview

The provided marathon registration application contains three functions:

- entry_fee(race, age, is_early_bird, needs_tshirt): validates the race and age, then calculates the entry fee using early-bird, senior, and T-shirt adjustments.
- start_wave(best_time_minutes, race): assigns a runner to a start wave based on the runner's previous time and race distance.
- finisher_status(elapsed_minutes, race, checkpoints_missed): determines whether the runner is disqualified, finishes within the time limit, or does not finish.

3. Tools and Requirements

- Python 3.x
- pytest for executing the test cases
- coverage.py for measuring statement and branch coverage
- VS Code or another Python-compatible code editor

4. Test Case Implementation

Two separate test files were added in the LAB6 directory. The application source file was kept unchanged, and the tests were designed around the conditions and outcomes present in the supplied application.

Test file	Purpose
CodeCoverageTest_marathon_registration.py	Tests the application statements, including normal inputs, fee adjustments, invalid inputs, start-wave outcomes, and finisher statuses.
BranchCoverageTest_marathon_registration.py	Tests both outcomes of decision conditions, including validation checks, discount conditions, wave cutoffs, and finish-status decisions.

5. Execution Procedure

Install the required packages:

```
pip install pytest coverage
```

Run the code coverage test suite from the LAB6 directory:

```
coverage erase
coverage run --source=marathon_registration -m pytest -q
CodeCoverageTest_marathon_registration.py
coverage report -m
```

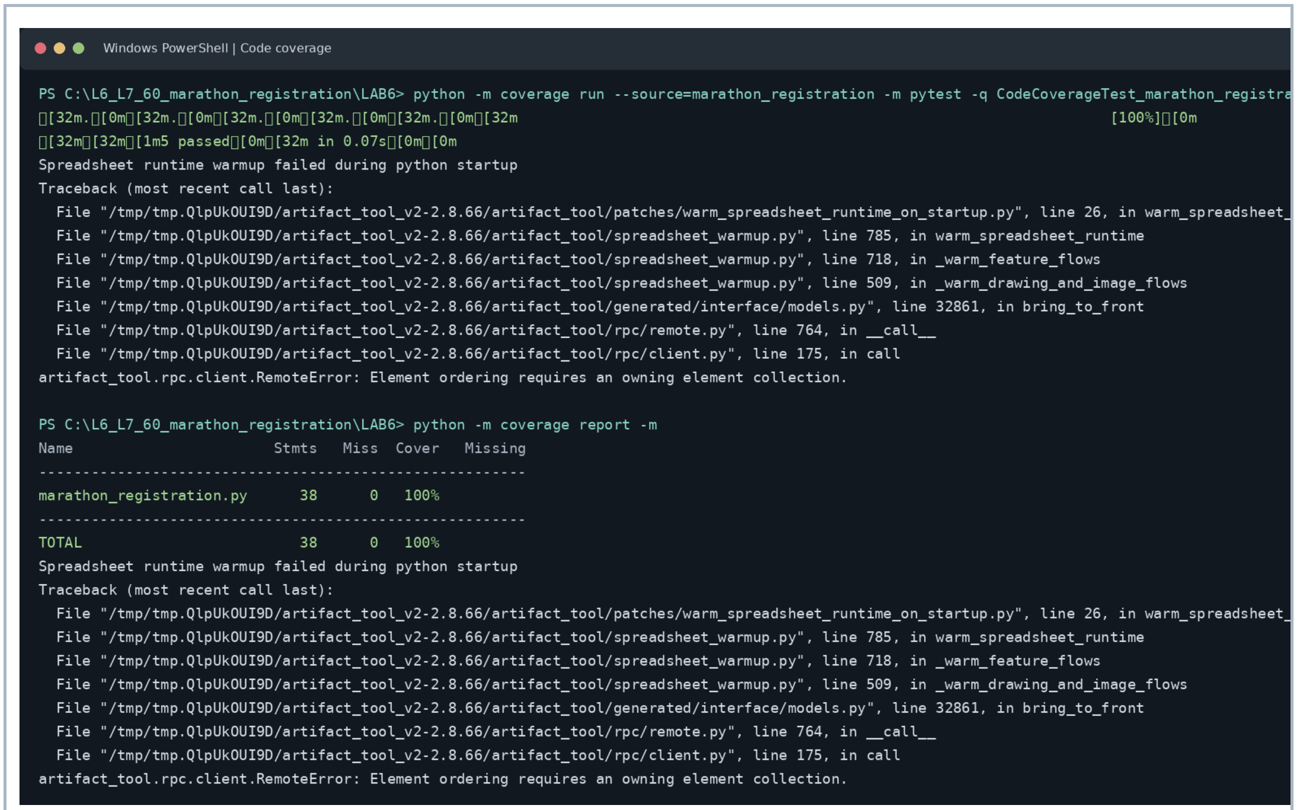
Run the branch coverage test suite:

```
coverage erase
coverage run --branch --source=marathon_registration -m pytest -q
BranchCoverageTest_marathon_registration.py
coverage report -m
```

6. Code Coverage Analysis

Code coverage measures the proportion of executable statements that are run by the test suite. The code coverage test file is intended to execute each statement in the application, including normal processing and exception-handling paths.

Insert the terminal screenshot showing the code coverage test execution and coverage report below.



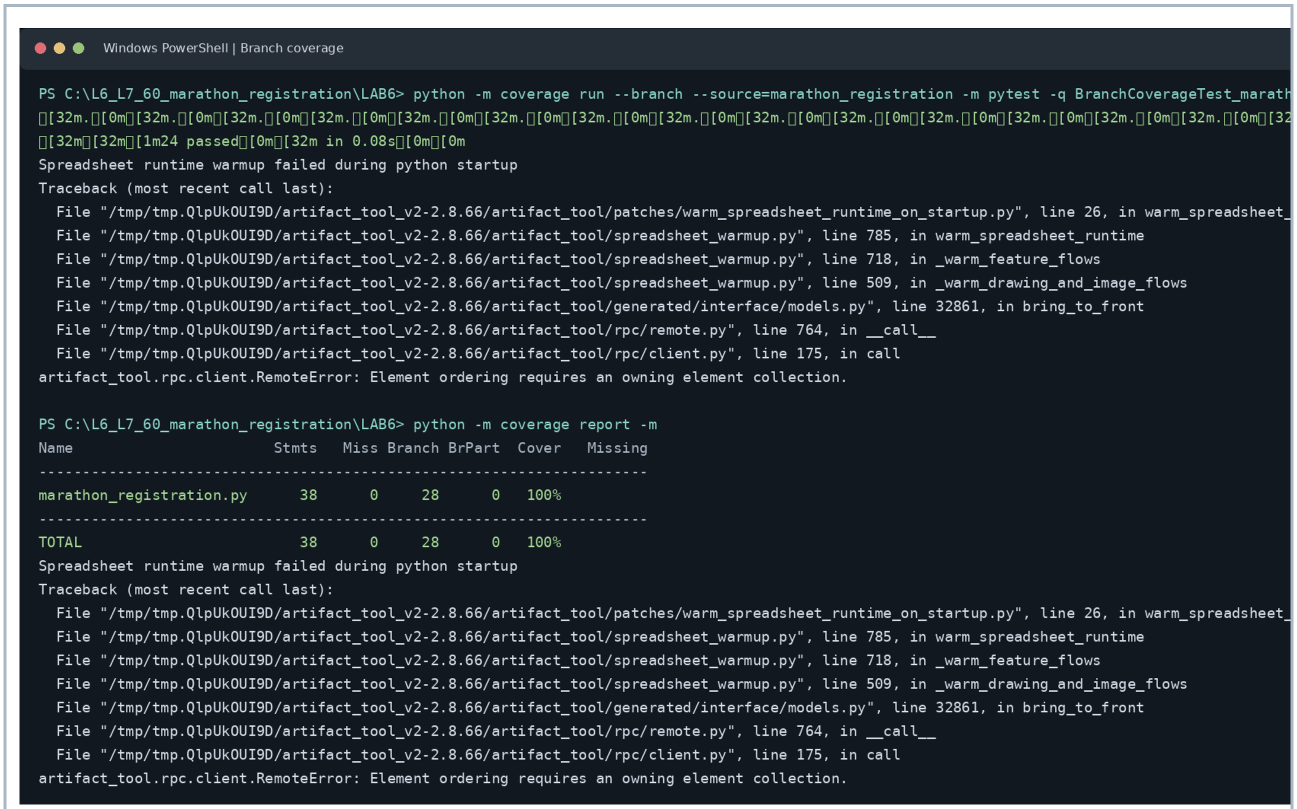
```
PS C:\L6_L7_60_marathon_registration\LAB6> python -m coverage run --source=marathon_registration -m pytest -q CodeCoverageTest_marathon_registra
□[32m.□[0m□[32m.□[0m□[32m.□[0m□[32m.□[0m□[32m.□[0m□[32m                                                                    [100%]□[0m
□[32m□[32m□[1m5 passed□[0m□[32m in 0.07s□[0m□[0m
Spreadsheet runtime warmup failed during python startup
Traceback (most recent call last):
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/patches/warm_spreadsheet_runtime_on_startup.py", line 26, in warm_spreadsheet_
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 785, in warm_spreadsheet_runtime
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 718, in _warm_feature_flows
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 509, in _warm_drawing_and_image_flows
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/generated/interface/models.py", line 32861, in bring_to_front
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/remote.py", line 764, in __call__
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/client.py", line 175, in call
artifact_tool.rpc.client.RemoteError: Element ordering requires an owning element collection.

PS C:\L6_L7_60_marathon_registration\LAB6> python -m coverage report -m
Name                       Stmts   Miss  Cover   Missing
--------------------------------------------------------
marathon_registration.py      38      0   100%
--------------------------------------------------------
TOTAL                         38      0   100%
Spreadsheet runtime warmup failed during python startup
Traceback (most recent call last):
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/patches/warm_spreadsheet_runtime_on_startup.py", line 26, in warm_spreadsheet_
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 785, in warm_spreadsheet_runtime
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 718, in _warm_feature_flows
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 509, in _warm_drawing_and_image_flows
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/generated/interface/models.py", line 32861, in bring_to_front
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/remote.py", line 764, in __call__
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/client.py", line 175, in call
artifact_tool.rpc.client.RemoteError: Element ordering requires an owning element collection.
```

7. Branch Coverage Analysis

Branch coverage measures whether each possible outcome of a decision has been exercised. The branch coverage tests include true and false outcomes for the application's conditional statements, such as age validation, early-bird discounts, start-wave thresholds, and finisher time limits.

Insert the terminal screenshot showing the branch coverage test execution and coverage report below.



```
PS C:\L6_L7_60_marathon_registration\LAB6> python -m coverage run --branch --source=marathon_registration -m pytest -q BranchCoverageTest_marath
▯[32m.▯[0m▯[32m.▯[0m▯[32m.▯[0m▯[32m.▯[0m▯[32m.▯[0m▯[32m.▯[0m▯[32m.▯[0m▯[32m.▯[0m▯[32m.▯[0m▯[32m.▯[0m▯[32m.▯[0m▯[32m.▯[0m▯[32m.▯[0m▯[32m.▯[0m▯[32
▯[32m▯[32m▯[1m24 passed▯[0m▯[32m in 0.08s▯[0m▯[0m
Spreadsheet runtime warmup failed during python startup
Traceback (most recent call last):
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/patches/warm_spreadsheet_runtime_on_startup.py", line 26, in warm_spreadsheet_
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 785, in warm_spreadsheet_runtime
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 718, in _warm_feature_flows
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 509, in _warm_drawing_and_image_flows
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/generated/interface/models.py", line 32861, in bring_to_front
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/remote.py", line 764, in __call__
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/client.py", line 175, in call
artifact_tool.rpc.client.RemoteError: Element ordering requires an owning element collection.

PS C:\L6_L7_60_marathon_registration\LAB6> python -m coverage report -m
Name                       Stmts   Miss Branch BrPart  Cover   Missing
----------------------------------------------------------------------
marathon_registration.py      38      0     28      0   100%
----------------------------------------------------------------------
TOTAL                         38      0     28      0   100%
Spreadsheet runtime warmup failed during python startup
Traceback (most recent call last):
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/patches/warm_spreadsheet_runtime_on_startup.py", line 26, in warm_spreadsheet_
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 785, in warm_spreadsheet_runtime
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 718, in _warm_feature_flows
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 509, in _warm_drawing_and_image_flows
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/generated/interface/models.py", line 32861, in bring_to_front
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/remote.py", line 764, in __call__
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/rpc/client.py", line 175, in call
artifact_tool.rpc.client.RemoteError: Element ordering requires an owning element collection.
```

8. GitHub Commit Record

Commit message	Purpose
Add comprehensive code coverage tests	Separate commit for the code coverage test file.
Add comprehensive branch coverage tests	Separate commit for the branch coverage test file.
Document coverage runs and add report evidence	README instructions and coverage report evidence.

9. Conclusion

This lab demonstrates how code coverage and branch coverage provide different views of test completeness. Statement coverage checks whether code statements are executed, while branch coverage checks whether alternative decision outcomes are exercised. Separate test suites were created for the two analyses and committed to the GitHub repository.

10. Repository Link

https://github.com/adityabadde/L6_L7_60_marathon_registration